

Repository Activity Analysis: scottyab/rootbeer

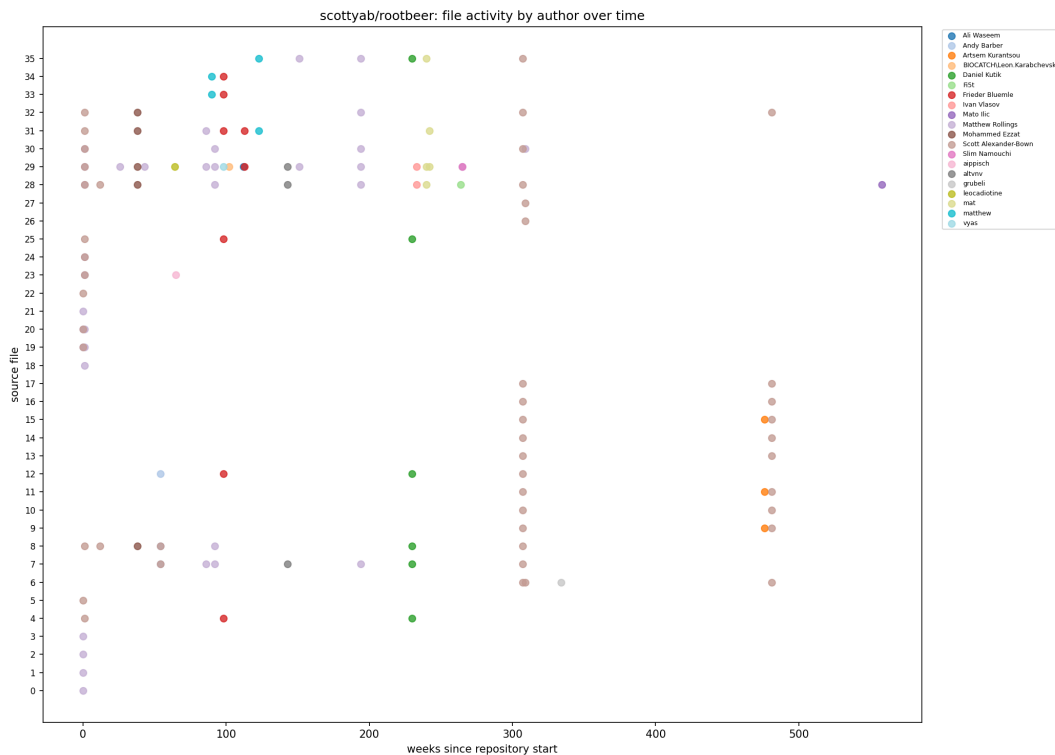
Analysis Overview

This report looks at development activity on the scottyab/rootbeer repository, an open-source Android library used to detect rooted devices. Source files were identified using GitHub's language breakdown for the repo (Java, Kotlin, C++, C), which came out to 36 source files total. For each of those files, I collected who changed it, when, and how many times, using GitHub's commit history. Merge commits were excluded so the same change wasn't counted twice. This produced 172 file-touch records covering the repository's full history.

Significant Findings

- 1. A small number of files get most of the attention.** RootBeer.java, the core detection logic, was touched 35 times, more than double the next file, Const.java (17 touches). The top five files together make up 80 of the 172 total touches, about 47%. Most of the codebase doesn't get touched nearly as often as this small core.
- 2. Most of the work has come from just two people.** Scott Alexander-Bown (61 touches) and Matthew Rollings (50 touches) account for 111 of the 172 touches, about 65% combined. If either of them stepped away from the project, a lot of institutional knowledge about the core logic would go with them.
- 3. Activity comes in bursts, not a steady stream.** The plot shows a lot of activity early on (roughly weeks 0-150), then another cluster around weeks 230-240, then a final burst around week 480, with long quiet stretches in between. This looks more like periodic maintenance pushes than ongoing day-to-day development.

Visualization



Each point is one author changing one source file during one week. The x-axis is weeks since the repo's first commit; the y-axis lists the 36 source files (shown as numbers instead of names to keep the plot readable); color shows the author. A row with points spread across a wide range of weeks, like RootBeer.java, means that file has been under long-term, ongoing maintenance. A row with points only near week 0 means that file was touched early and mostly left alone since.

Limitation

I defined a source file as one written in the repo's main compiled languages (Java, Kotlin, C++, C, and C/C++ headers), based on GitHub's language breakdown. This leaves out other files in the repo, like XML files, that I didn't count as source but that could still be part of the picture. A different definition of "source file" might show a different amount of contributor activity than what's shown here.